

An Occupancy-Indexed Operating Envelope for Compact Physics-Guided Automatic Modulation Classification under Structured Interference

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Abstract—Uniform model rankings hide a central design question in automatic modulation classification (AMC): when does a compact physics-guided representation beat a much larger I/Q-domain classifier under structured interference? We study this question in source-disjoint 3GPP TR 38.901 TDL-profile simulations using VIMD-Net, a 39500-parameter spectral network with a simulation-component teacher, three soft allocation routes, and exact-source cross-condition training. A two-variable empirical envelope fitted only on in-distribution and hard-interference cells relates its gain to jammer spectral occupancy and signal-to-interference ratio (SIR). Against an IQFormer-inspired reference, the envelope attains 0.975 sign agreement over 81 cells from unseen regimes. At $\text{SIR} = -15$ dB, the compact model gains 9.74 percentage points over MCLDNN and 10.46 percentage points over IQFormer-inspired, with positive paired intervals and the same direction in every seed. The sealed confirmatory method effect over the shared backbone is 4.57 percentage points with a simultaneous interval of $[3.65, 5.49]$ percentage points; however, the confirmatory family gate and the clean-retention gate did not pass. Frozen-checkpoint probes and prospective Tier-2 experiments localize the clean boundary to a repairable representation gap: adding a 46794-parameter received-I/Q sidecar improves clean macro-F1 by 11.77 percentage points and hard-interference macro-F1 by 4.37 percentage points. These Tier-2 results are exploratory, outside the frozen confirmatory family. The paper therefore reports a simulation-only operating envelope and a repair path, not uniform superiority.

Index Terms—Automatic modulation classification, structured interference, vehicular Doppler, operating envelope, representation allocation, compact neural networks.

Evidence status. The sealed primary method contrast is positive, but the confirmatory family gate and clean-retention gate did not pass. Post-hoc stratification is explicitly labeled. Tier-2 probe, repair, and capacity results are exploratory and outside the frozen confirmatory family. Every numerical value is generated from a source artifact; the scope is simulation only.

I. INTRODUCTION

AUTOMATIC modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform’s modulation from received samples. Deep AMC has progressed from convolutional I/Q classifiers [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], but average accuracy alone does not say where a compact representation is preferable.

Structured interference makes that omission consequential. A sweep, tone, pulse, partial-band process, or modulated

out-of-taxonomy emitter may overlap the target in time and frequency. The same model can trail a high-capacity I/Q classifier in weak or narrowband interference yet lead it in a severe, spectrally occupied corner. Averaging these regimes produces a ranking but erases the design rule.

This paper asks a narrower question: *can the sign and size of the compact-model advantage be indexed by physical condition variables, and can the boundary revealed by that index be repaired?* We answer using a frozen source-disjoint simulation campaign, artifact-only post-hoc analysis, and prospective Tier-2 experiments. The base model, VIMD-Net, maps a complex STFT to target-dominant, jammer-dominant, and unexplained-or-ambiguous routes. Its teacher is available only during simulation training and is derived from independently tracked target, jammer, and residual components.

The contributions are:

- an occupancy–SIR operating envelope that unifies gains and losses and predicts their sign outside the regimes used to fit it;
- a statistically reinforced severe-interference result in which the 39500-parameter model exceeds 354984–406070-parameter baselines, together with full-region maps that prevent selective reporting;
- a sealed confirmatory backbone-relative effect and an honest bounded conclusion for component-level interventions whose joint family gate did not pass;
- a condition-transfer diagnosis and prospective repair chain showing that clean exposure alone is insufficient, while a small received-I/Q sidecar repairs both clean and hard-interference performance; and
- a traceable evidence layer in which manuscript numbers and figures resolve to immutable artifact files and selectors.

The claim is deliberately not that VIMD-Net is uniformly best. It is that compact physics-guided spectral allocation has a reproducible operating region, that the region can be predicted from physical covariates, and that its principal clean boundary is architectural rather than an unavoidable cost of compactness.

II. RELATED WORK AND CLAIM BOUNDARY

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Channel adaptation, masked or contrastive learning, and interference-tolerant features address different

nuisance sources [9]–[12]. These papers motivate strong I/Q-domain comparators and prevent us from presenting masking or contrastive learning alone as novelty.

The 2024–2026 frontier also includes overlapped-signal classification, denoising masked autoencoding, cross-domain prototypical adaptation, and a tensor method for malicious interference in MIMO systems [13]–[16]. Their signal models and information contracts are not interchangeable with our frozen jammer-factor protocol, so we cite them for positioning rather than importing published numbers into the unified-retraining table. Accordingly, our claim is an audited operating boundary and repair chain under controlled simulation, not universal state of the art.

Time–frequency masks are established separation targets. Binary, ratio, phase-sensitive, complex-ratio, and deep-clustering formulations have been studied extensively [17]–[21]. Our use differs in purpose: the mask allocates evidence for classification and does not reconstruct sources. The third route retains unexplained power and target–jammer ambiguity that a suppression system might otherwise discard.

The main distinction of this work is evaluative. Rather than infer uniform superiority from a marginal score, we compare compact spectral and larger I/Q representations across an explicit physical plane, estimate an empirical boundary, test it on held-out regimes, and then intervene on the boundary. All evidence is restricted to TR 38.901 TDL-profile simulations. No claim is made about complete V2X geometry, waveform recovery, source separation, open-set recognition, or hardware deployment.

III. SYSTEM MODEL AND METHOD

A. Received Signal and Paired Views

For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \gamma \mathcal{H}_u(u)[n] + w[n]), \quad (1)$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators, γ controls SIR, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. Training pairs two independently impaired observations of the same immutable source. Inference uses one view. Pairing is therefore a training and statistical-alignment device, not additional test-time information.

Unanchored in-taxonomy cochannel collisions are excluded. Without a physical target anchor, exchanging two admissible emitters can preserve the mixture while changing the requested single label, so that task requires a different label definition.

B. Three-Route Representation Allocation

Let $Z = \text{STFT}(x) \in \mathbb{C}^{F \times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned} [M_s, M_j, M_o] &= \text{softmax}([A_s, A_j, A_o]), \\ M_s + M_j + M_o &= 1. \end{aligned} \quad (2)$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ . The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \quad (3)$$

$$Z_j = (M_j + (1 - \lambda)M_o) \odot Z. \quad (4)$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. This elementary property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student’s lattice. With $q_k = P_k / (P_s + P_j + P_u)$, the fixed teacher is

$$M_s^* = [q_s - q_j]_+, \quad M_j^* = [q_j - q_s]_+, \quad (5)$$

$$M_o^* = q_u + 2 \min(q_s, q_j). \quad (6)$$

The routes are nonnegative and sum to one. Common scaling cancels in the ratios. Negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. Jensen–Shannon supervision aligns student and teacher masks; modulation, jammer-family, and link-quality objectives constrain route semantics; and an exact-source contrastive loss encourages condition invariance. The teacher is not needed at inference.

IV. EVIDENCE PROTOCOL

A. Frozen Simulation Campaign

The artifact-locked campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes cover PSK, QAM, continuous-phase, and FSK families. Structured jammer families span tone, multitone, chirp, partial band, pulse, broadband, and out-of-taxonomy modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons.

Comparators include a shared direct spectral backbone (A0), MCLDNN, an IQFormer-inspired classifier, and an official-architecture CSSL supervised adaptation. The full method is A5. Ablations alter route count, teacher form, multi-task terms, contrastive learning, and the residual bypass. Strong comparators are anchors, not claims of exact reproduction of every original training recipe.

B. Statistical Separation of Evidence Classes

The sealed confirmatory family contains three hard-interference contrasts: the complete method versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. A joint non-studentized max-absolute-deviation hierarchical paired bootstrap supplies simultaneous intervals over seeds and class-stratified source clusters. The family gate requires every simultaneous lower bound to exceed zero.

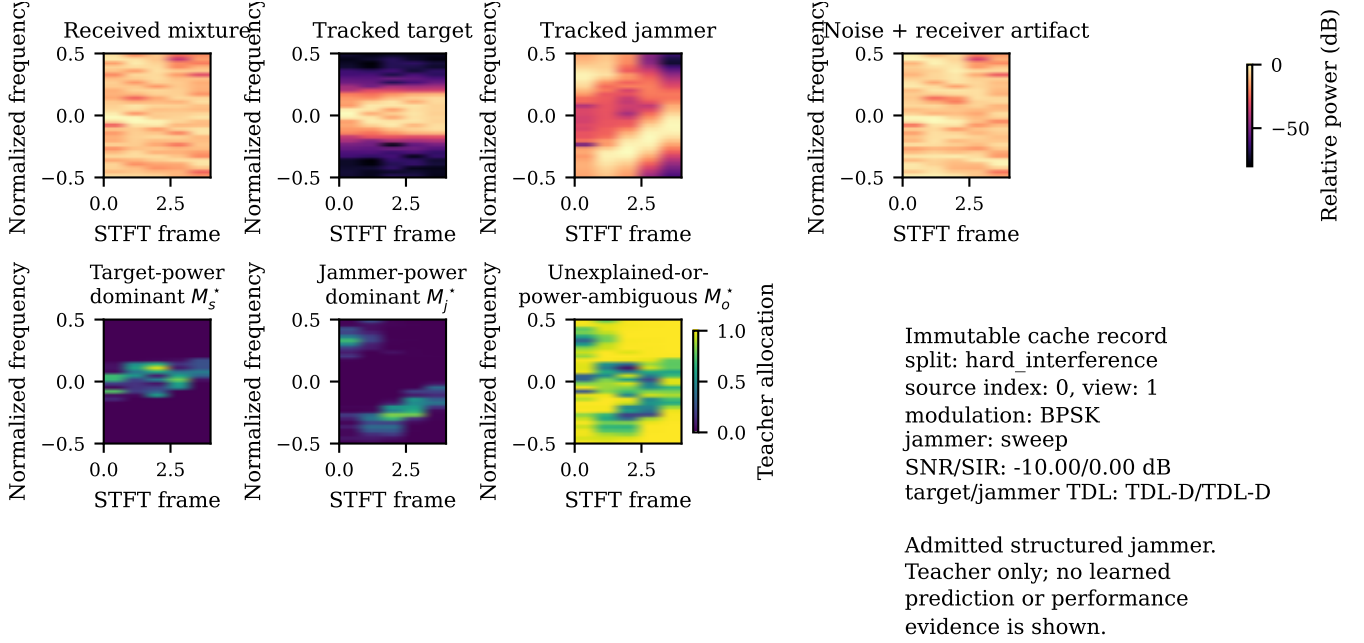


Fig. 1. A fixed-teacher proxy visualization from an immutable simulation record. The upper row shows the received mixture and tracked components; the lower row shows the three allocations on the same STFT lattice. The image explains the teacher definition and is not performance evidence.

All SIR-cell, occupancy, fusion, condition-transfer, probe, repair, and capacity analyses are exploratory. In particular, the severe SIR stratification is post hoc. Its robustness checks include the complete SNR-SIR map, multiplicity correction over inspected cells, source/seed bootstrap variants, and a sign-flip test. Tier-2 experiments were designed after inspecting the frozen evidence and are outside the frozen confirmatory family; they cannot retroactively change a sealed conclusion.

The clean-retention gate was preregistered separately. It did not pass. Consequently, clean behavior is treated as a boundary to explain and repair, not as a preserved property of the sealed model.

V. RESULTS

A. Confirmed Whole-Method Effect, Bounded Component Effects

Table I reports the complete confirmatory family. The whole method improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of [3.65, 5.49] percentage points. This effect is about three times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their contributions are bounded within 1.36 percentage points. Thus the architecture-level effect is resolved but cannot be assigned to either isolated component at this resolution. Because the family rule required all three contrasts, the family gate did not pass.

At the same parameter tier, the residual-free A7 variant is nominally above A5 in the marginal hard-interference aggregate (Fig. 5). The difference is well below the design resolution and is consistent with the bounded component-level result. We therefore attribute the compact frontier position to the VIMD family at this budget, not uniquely to the bypass.

TABLE I
SEALED CONFIRMATORY HARD-INTERFERENCE FAMILY. DIFFERENCES ARE CANDIDATE MINUS REFERENCE IN MACRO-F1 PERCENTAGE POINTS.

Contrast	Diff.	Sim. low	Sim. high
Full method vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	-0.76	1.08
Tri-route vs dual-route	0.44	-0.47	1.36

B. Rank Exchange across the Interference Plane

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired. Those averages favor the larger models, but Fig. 2 shows why no single ranking is adequate. Rank changes with SNR and SIR.

At SIR = -15 dB, A5 gains 9.74 percentage points over MCLDNN ([7.52, 12.14]) and 10.46 percentage points over IQFormer-inspired ([6.78, 14.29]). All 10 seeds have the same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. These checks strengthen the corner result without converting the post-hoc analysis into confirmation.

C. Occupancy-Indexed Envelope and Out-of-Regime Test

The rank exchange motivates an empirical law rather than another marginal average. For reference b , we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b}o + \beta_{2,b}\text{SIR}, \quad (7)$$

where o is jammer spectral occupancy and Δ_b is A5 minus the reference macro-F1. Only ID and hard-interference cells enter the fit.

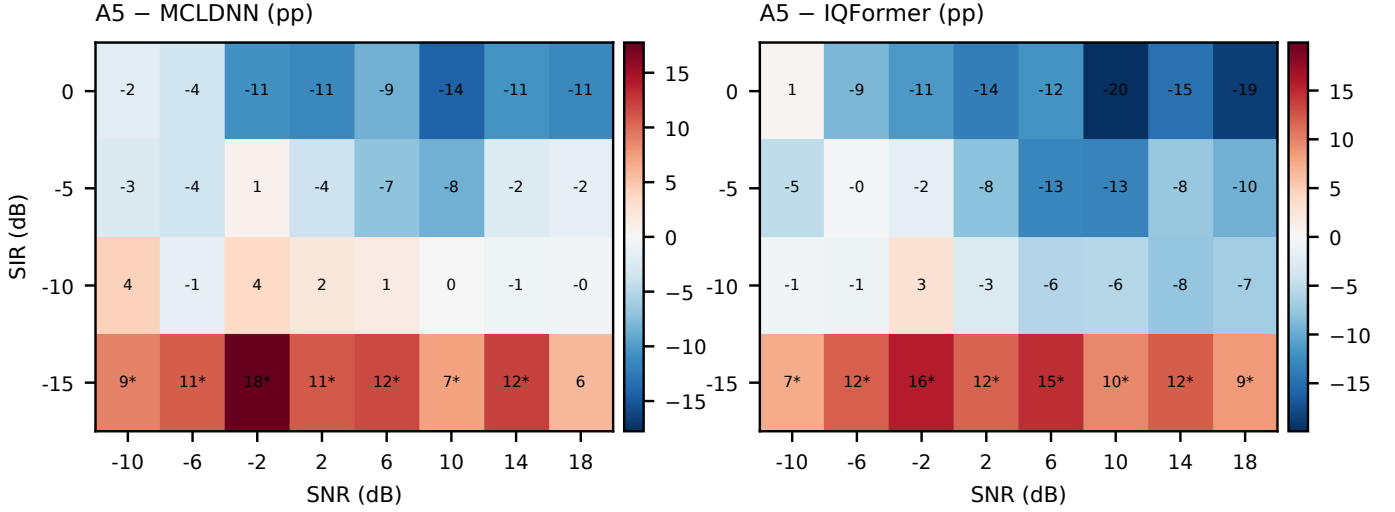


Fig. 2. Post-hoc operating map over all hard-interference SNR–SIR cells. Each entry is A5 minus the named strong baseline in macro-F1 percentage points; a star marks a positive lower paired interval. Displaying the full map prevents the severe corner from being read as a uniform ranking.

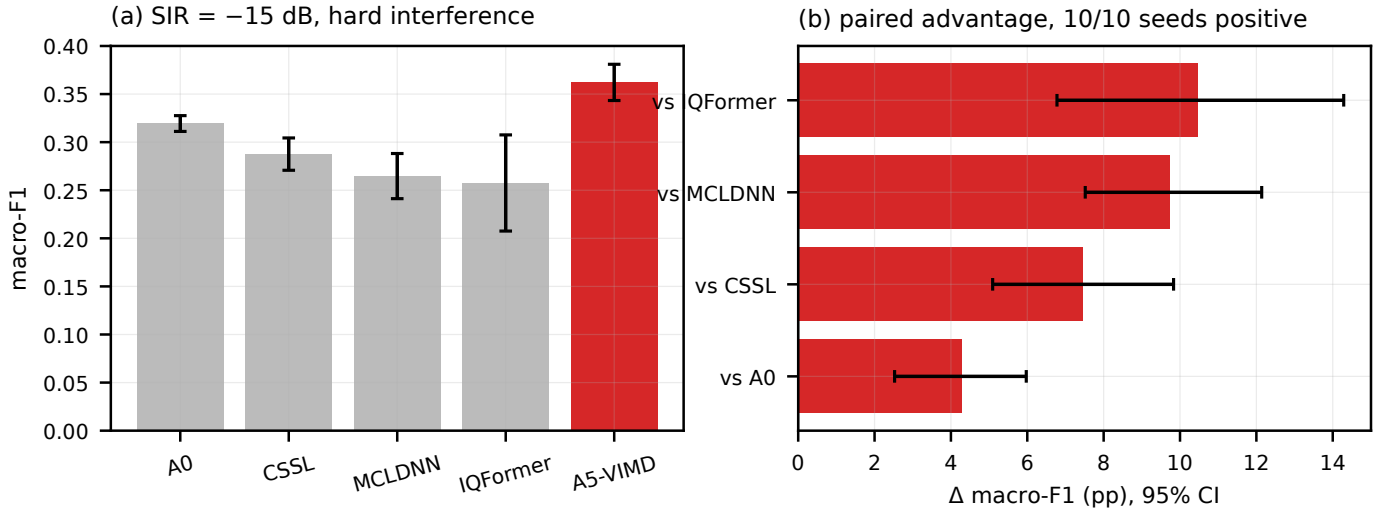


Fig. 3. Severe-interference corner. Left: absolute macro-F1 at $\text{SIR} = -15$ dB. Right: source-paired A5 advantages with stratified bootstrap intervals. The comparison is exploratory and post hoc.

Against IQFormer-inspired, the fitted occupancy coefficient is 11.39 percentage points per unit occupancy and the SIR coefficient is -0.666 percentage points per dB. On 81 cells from five unused regimes, the law obtains Pearson $r = 0.694$, sign agreement 0.975, and RMSE 6.20 percentage points. The MCLDNN envelope gives sign agreement 0.951. Thus losses on clean and held-jammer conditions and gains in severe occupied interference are evaluations of one low-dimensional boundary, not unrelated exceptions.

Solving Eq. (7) for zero gain yields the break-even occupancies in Table II. The required occupancy rises as interference becomes less severe, making the boundary directly usable as a model-selection diagnostic.

The preregistered mechanism expectation was non-increasing gain with occupancy. The sealed directional test rejected that sign. Exploratory quintiles instead show monotonically increasing A5–CSSL gain, from a negative low-occupancy region to a strongly positive high-occupancy region.

TABLE II
EXPLORATORY BREAK-EVEN JAMMER SPECTRAL OCCUPANCY.

SIR (dB)	vs IQFormer	vs MCLDNN
−15	0.58	0.48
−10	0.87	0.66

Because occupancy and SIR co-vary in this design, Eq. (7) is predictive and descriptive, not a causal structural equation.

D. Complementarity, Cost, and Selective Prediction

A5 remains useful outside its winning region because its errors differ from those of the I/Q models. Probability fusion with IQFormer-inspired improves over the stronger constituent by 4.71 percentage points, whereas a same-architecture seed-ensemble control improves by only 1.49 percentage points. The corresponding MCLDNN fusion/control gains are

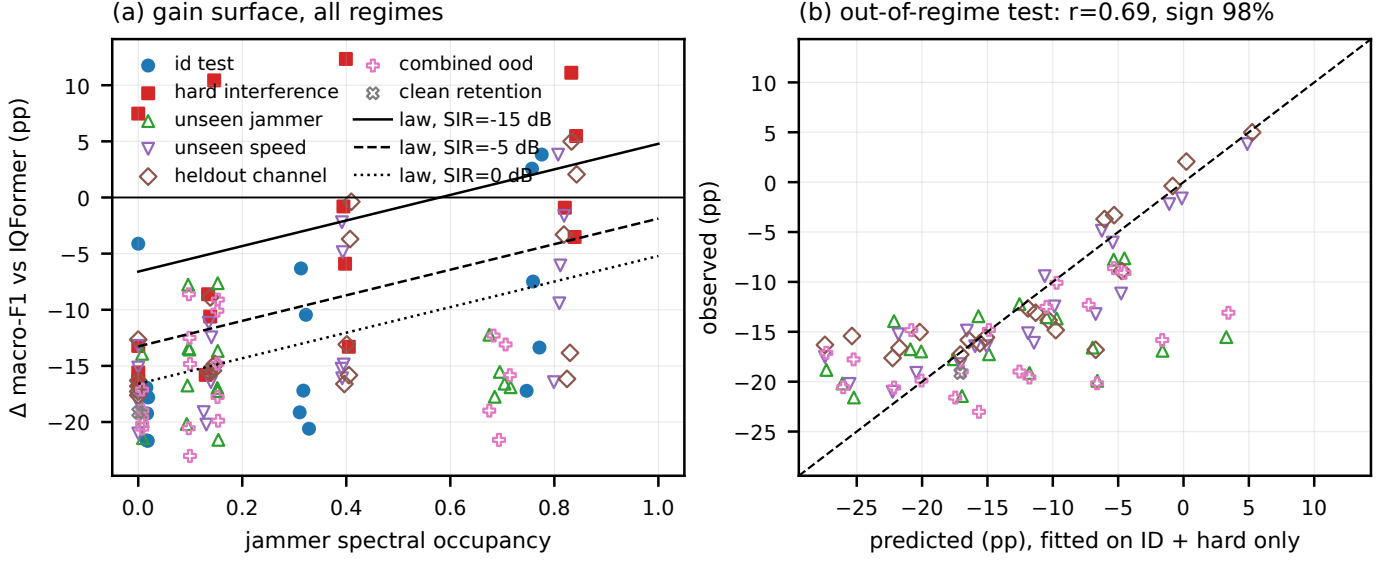


Fig. 4. Exploratory occupancy-SIR envelope. Filled cells fit the law; open cells are unused regimes. The right panel compares out-of-regime predictions with artifact-derived outcomes.

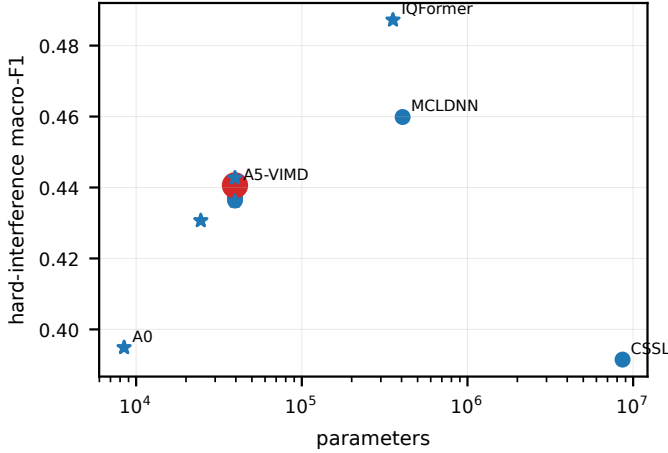


Fig. 5. Hard-interference macro-F1 versus parameter count. Stars mark the artifact-derived Pareto set. The frontier is regime-specific.

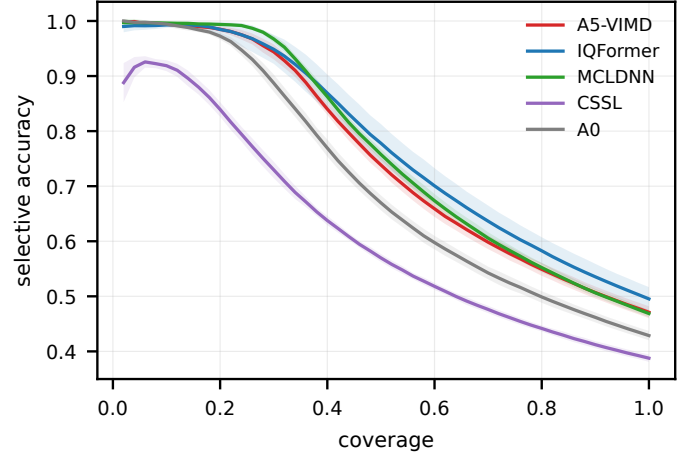


Fig. 6. Selective accuracy versus coverage on hard interference. Thresholds are calibrated without using test labels.

5.70/1.69 percentage points. This is evidence of complementary decisions, not evidence that the compact model alone wins the marginal benchmark.

A5 uses 39500 parameters and 41.8 million reported MACs, versus 354984 and 355.6 million for IQFormer-inspired. At 30% retained coverage after validation-only confidence calibration, A5 achieves 94.4% selective accuracy, close to the IQFormer-inspired 94.9%. Figure 6 shows the full risk-coverage curves rather than one selected point.

E. Clean Boundary: Coverage, Representation, and Repair

The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes; the remaining 6 classes are a condition-transfer test. Among uncovered constellation-order cells, transfer failure occurs in 100% of compact-spectral cells versus 11% of I/Q high-capacity cells. The clean-retention gate did not pass.

Figure 7 localizes the failure rather than hiding it in an aggregate.

We then followed a prospective decision chain (Table III). Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation. For A5, 5 of 6 design-seed cells lie below the preregistered high-order threshold 0.25; one cell is borderline. The strong I/Q baselines are above that threshold in 18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points ($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient.

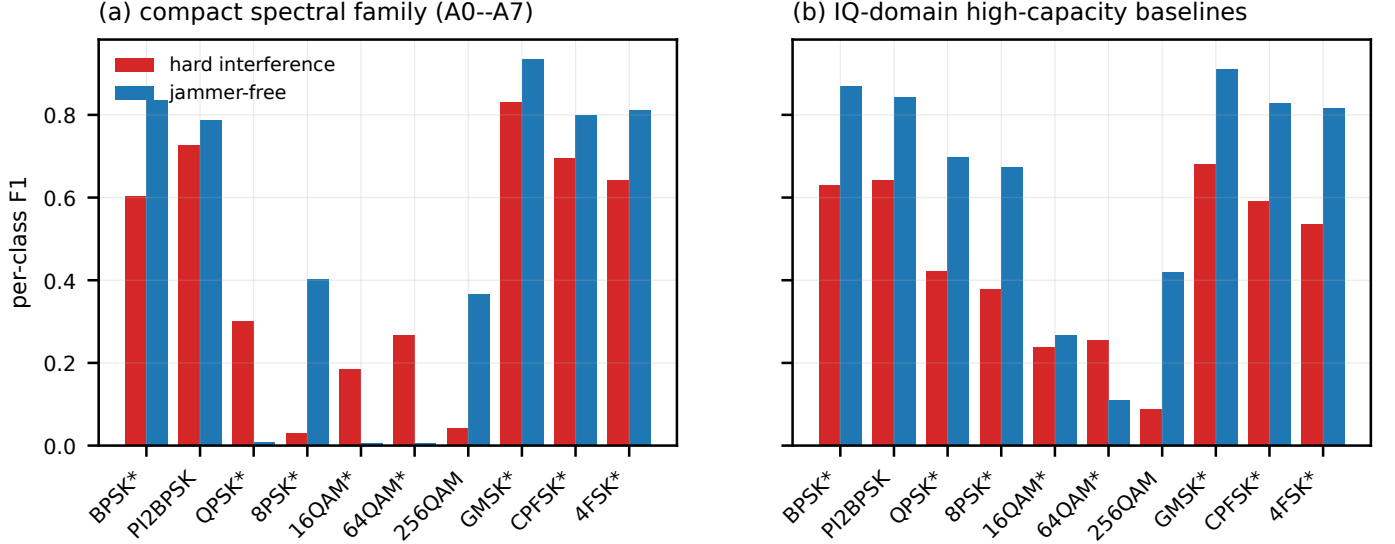


Fig. 7. Condition-transfer taxonomy. Training coverage and discriminability type separate the compact spectral failure from classes whose envelope cues remain distinctive.

A preregistered negative control gates the modulation route using predicted interference presence. Relative to A5, it changes clean macro-F1 by -0.30 percentage points $[-0.66, 0.08]$ and hard macro-F1 by -0.18 percentage points $[-0.72, 0.35]$. The learned gate averages 99.994% on clean windows and 99.991% under hard interference, so it saturates near the modulated route in both conditions rather than implementing the intended clean identity fallback. This arm does not repair clean performance; it is a negative control, not stand-alone evidence that a functional mask fallback was successfully learned.

The next intervention adds a 46794-parameter received-I/Q sidecar while retaining the spectral routes and the same source mixture. It raises clean macro-F1 by 11.77 percentage points $[11.06, 12.49]$ and hard macro-F1 by 4.37 percentage points $[3.44, 5.31]$. Its overall hard score is statistically near IQFormer-inspired ($\Delta = -0.29$ percentage points, $[-1.95, 1.46]$), while at severe SIR it remains ahead by 12.09 percentage points $[8.32, 15.88]$. This repairs the clean boundary without erasing the structured-interference region.

F. Capacity Ladder Separates Size from Representation

Finally, we widen the spectral model without changing its representation domain. The S, M, and L tiers contain 39500, 99596, and 233244 parameters. Relative to S, M improves hard macro-F1 by 1.94 percentage points and L by 3.18 percentage points. L also improves over M by 1.24 percentage points. Capacity therefore improves average fitting.

Nevertheless, L remains below IQFormer-inspired on overall hard interference by -2.84 percentage points $[-4.30, -1.31]$. At $\text{SIR} = -15$ dB, the same L model is ahead by 14.07 percentage points $[9.55, 18.51]$. Increasing spectral capacity therefore shifts performance but does not remove the operating boundary. The result supports an architecture/representation interpretation rather than a pure parameter-count explanation.

VI. DISCUSSION

The evidence suggests a practical model-selection rule. A compact spectral front end is most attractive when interference is both severe and occupies enough of the time-frequency plane for learned allocation to matter. A larger I/Q model is preferable in weak, clean, or low-occupancy conditions unless the compact system includes an I/Q side channel. The two models are also complementary enough that late probability fusion can outperform either alone.

The sequence of interventions matters scientifically. The frozen probe narrows the cause; the coverage arm rejects a cheap explanation; the I/Q sidecar repairs the predicted missing information; and the capacity ladder shows that width alone does not eliminate the boundary. This chain is more informative than a single improved score because each negative or positive outcome discriminates between competing explanations.

The envelope is intentionally empirical. Occupancy and SIR are correlated in the current design, jammer occupancy is estimated from simulator-tracked components, and the law has only been tested on regimes contained in the frozen cache. Its value is as a compact prediction and audit device. A future prospective campaign should randomize occupancy and SIR more independently and estimate deployment-available occupancy proxies.

VII. LIMITATIONS AND REPRODUCIBILITY

The evidence has four explicit limits. First, all results are from source-disjoint TR 38.901 TDL-profile simulations, not a complete vehicular geometry. Second, the teacher uses simulation-component bookkeeping and cannot supervise un-referenced recordings. Third, the confirmatory family gate and clean-retention gate did not pass; only the complete A5-A0 contrast is a positive sealed confirmatory result. Fourth, the envelope, severe-cell, fusion, condition-transfer, probe, sidecar,

TABLE III
PROSPECTIVE TIER-2 DIAGNOSIS, REPAIR, AND NEGATIVE CONTROL. THESE RESULTS ARE EXPLORATORY AND OUTSIDE THE FROZEN CONFIRMATORY FAMILY.

Intervention	Question	Clean diff. [95% CI]	Hard diff. [95% CI]	Decision
Per-class clean coverage	Is exposure alone sufficient?	0.15 [-0.31,0.63]	-0.30 [-0.91,0.27]	No; clean gate misses by a wide margin.
Presence-gated route vs A5	Is mask application the primary clean cause?	-0.30 [-0.66,0.08]	-0.18 [-0.72,0.35]	No repair; $g \approx 1$ in both conditions.
Received-I/Q sidecar vs A5	Is the representation gap repairable?	11.77 [11.06,12.49]	4.37 [3.44,5.31]	Yes; both intervals are positive.

TABLE IV
EXPLORATORY CAPACITY LADDER ON HARD INTERFERENCE.

Contrast	Diff.	Low	High
M minus S	1.94	1.07	2.80
L minus S	3.18	2.44	3.96
L minus M	1.24	0.31	2.37
L minus IQFormer	-2.84	-4.30	-1.31

and capacity analyses are exploratory, with post-hoc status stated where applicable.

Reproducibility is enforced at the artifact boundary. Each manuscript macro stores the source path, row selector, evidence class, precision, and source SHA-256. Figures are rendered from CSV files and carry a separate provenance manifest. The frozen run stores cache identity, configurations, checkpoints, source-aligned predictions, seed aggregates, paired statistics, environment metadata, and execution records. These controls prevent a favorable Tier-2 result from overwriting an unfavorable sealed gate.

VIII. CONCLUSION

Compact physics-guided AMC and high-capacity I/Q AMC exchange rank across the structured-interference plane. An occupancy-SIR law predicts that exchange outside its fit regimes, and a post-hoc but statistically reinforced severe corner shows a reproducible advantage for the compact model. The sealed whole-method effect over a shared backbone is positive, while two isolated component effects are bounded and the confirmatory family gate does not pass. The clean-retention gate also does not pass, but the probe-coverage-sidecar chain localizes and repairs the representation gap; the capacity ladder shows that parameter count alone cannot explain the remaining boundary. The result is an honest simulation-only operating envelope with a concrete repair path, not a claim of uniform superiority.

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